THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · HARD LEVEL UNIT TEST SET 3

Chapters 5–8: **Measurement of Length and Motion, Materials Around Us, Temperature and its Measurement, A Journey through States of Water** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A clinical thermometer reads 36.8 °C and 37.0 °C in two successive tests. This suggests: 1

(a) hypothermia **(b)** error
(c) normal temperature with small variation **(d)** fever

2. A tank of 1000 L loses 2 L per day by evaporation. After 30 days, the water lost is: 1

(a) 30 L **(b)** 60 L **(c)** 600 L **(d)** 2 L

3. Which of these is the best reason for taking repeated measurements? 1

(a) to reduce random errors **(b)** to make the object longer
(c) to change units **(d)** to waste time

4. A bottle cap made of plastic is replaced by a metal cap. Which problem might arise? 1

(a) it may rust or conduct heat **(b)** it dissolves
(c) it becomes transparent **(d)** it floats

5. Which of the following devices uses an insulator to protect the user? 1

(a) iron rod **(b)** electric iron handle
(c) kettle body **(d)** frying pan base

6. Which of the following changes is reversible? 1

(a) cooking of rice **(b)** rusting **(c)** melting of ice **(d)** burning of wood

7. A thread of length 188 cm goes exactly 4 times round the edge of a circular plate. The circumference of the plate is: 1

(a) 47 cm **(b)** 188 cm **(c)** 94 cm **(d)** 752 cm

8. Which of these shows that solubility depends on the substance and not only on appearance? 1

(a) sugar and chalk powder both look white but behave differently in water
(b) both are white
(c) both are powders
(d) both are solid

9. A student records water temperature in the evening: 20 °C, 20 °C, 20 °C after placing in a room at 20 °C. This shows: 1

(a) the water is being heated
(b) the water is at the same temperature as the room
(c) the water is cooling
(d) the water is boiling

10. If the surface area of a liquid is doubled, the rate of evaporation will: 1

(a) increase **(b)** stay the same **(c)** stop **(d)** decrease

11. Which statement about a ceiling fan switched on is correct? 1

(a) the blades move randomly **(b)** the fan is at rest
(c) each blade moves linearly **(d)** each blade moves in a circular path

12. After removing iron filings, adding water to the mixture and filtering separates: 1

(a) salt on the filter paper **(b)** sand on the filter paper
(c) both on the filter **(d)** nothing

13. **Assertion (A):** A clinical thermometer is used to measure the temperature of a human body. 1
Reason (R): A laboratory thermometer is used to measure the temperature of boiling water.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. **Assertion (A):** Wet clothes dry faster on a windy day. 1
Reason (R): Moving air carries away water vapour quickly, so more water can evaporate.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

15. **Assertion (A):** The motion of a swing is periodic. 1
Reason (R): A swing repeats its motion after equal intervals of time.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Which has more heat: a bucket of water at 40 °C or a cup at 90 °C? Explain. 2

17. How would water table levels change with excessive pumping? 2

18. Why can a circle's circumference not be measured with a straight ruler directly? 2

19. Why does a metal spoon in a cup of hot tea become hot but a plastic one hardly changes? 2

20. A metal spoon and a wooden spoon at room temperature feel different to touch. Does this mean they have different temperatures? 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Plan a fair test to find which material keeps water hot longest: wool, cotton, aluminium foil. 3

22. Explain with an experiment how to prove that the water which disappears from a pond goes to the air. 3

23. Explain, with two examples each, the difference between rotation and revolution. 3

24. Give three reasons why plastic has replaced many materials in daily life. 3

OR

Explain how a daily temperature chart can help predict the next day's weather (basic idea). 3

25. Describe how an igloo can survive in cold weather and not melt. 3

OR

Why do we say that a train passenger and platform observer disagree about speed? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. **Case study.** Ananya in Hyderabad felt hot and weak. Her father took a clinical thermometer, shook it so that the mercury level came below 35 degrees C, and placed it under her tongue for about a minute. The reading was 38.5 degrees C. 5

(i) Why was the thermometer shaken first?

(ii) What is the normal body temperature?

(iii) In which unit is the temperature read here?

(iv) Does Ananya have a fever? Explain why the thermometer should not be held in hot water.

27. Explain how heat is transferred in a metal spoon placed in boiling soup and why the plastic spoon stays cool. 5

OR

Explain how a solar still can convert sea water to drinking water, naming the changes of state. 5

END OF QUESTION PAPER — Total: 50 Marks

MASTERMIND